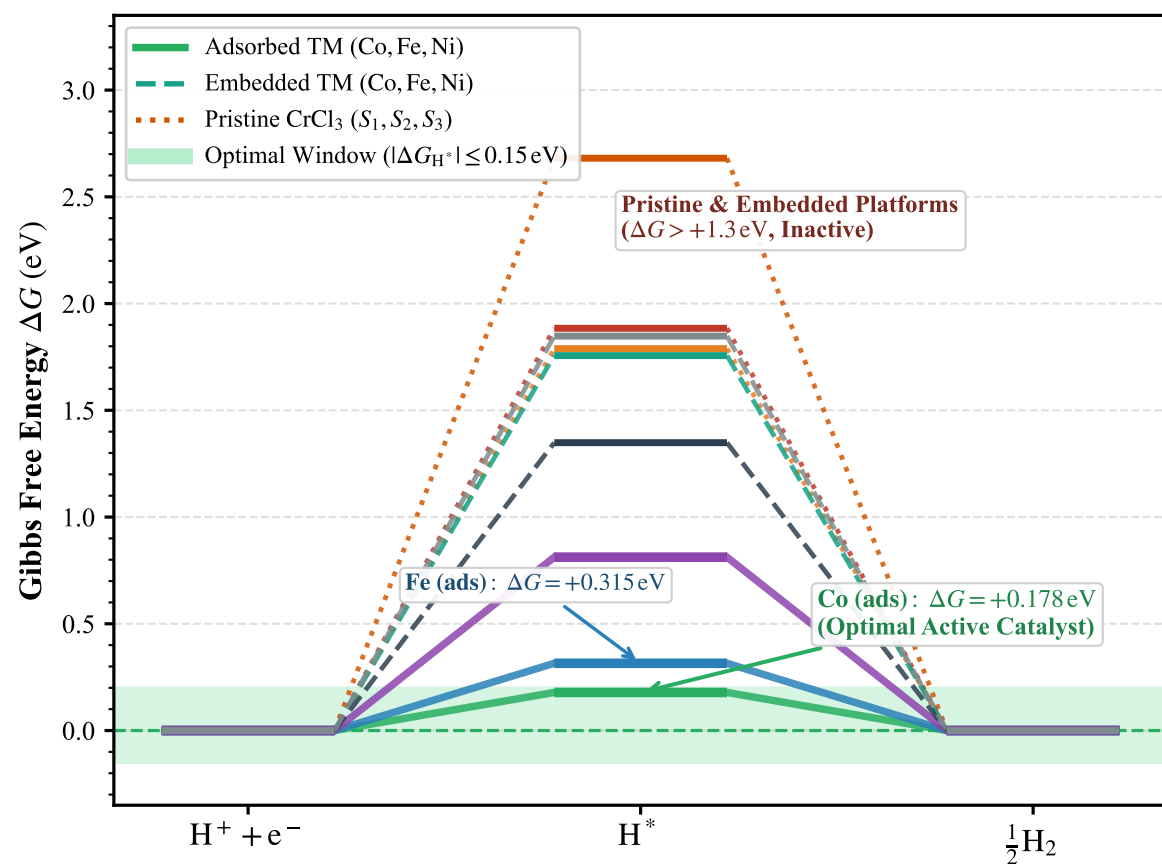
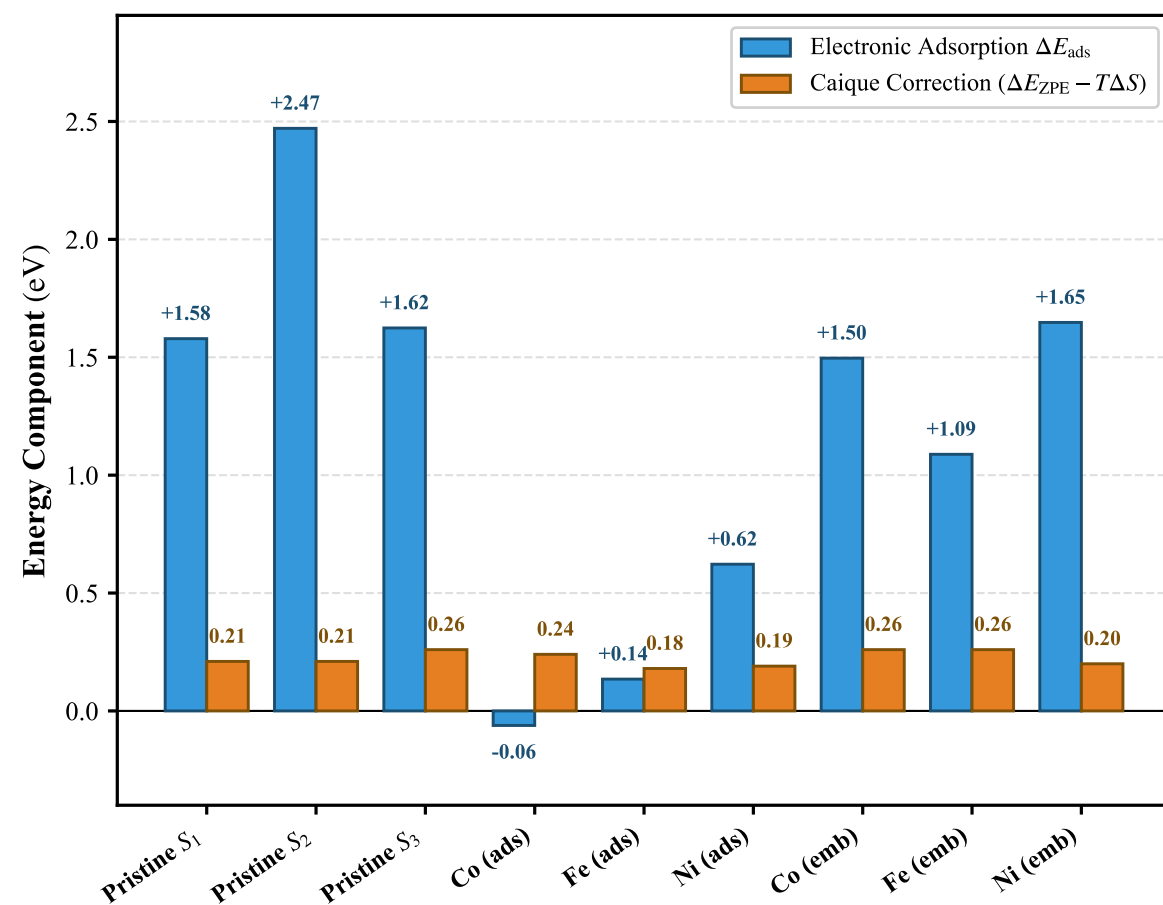
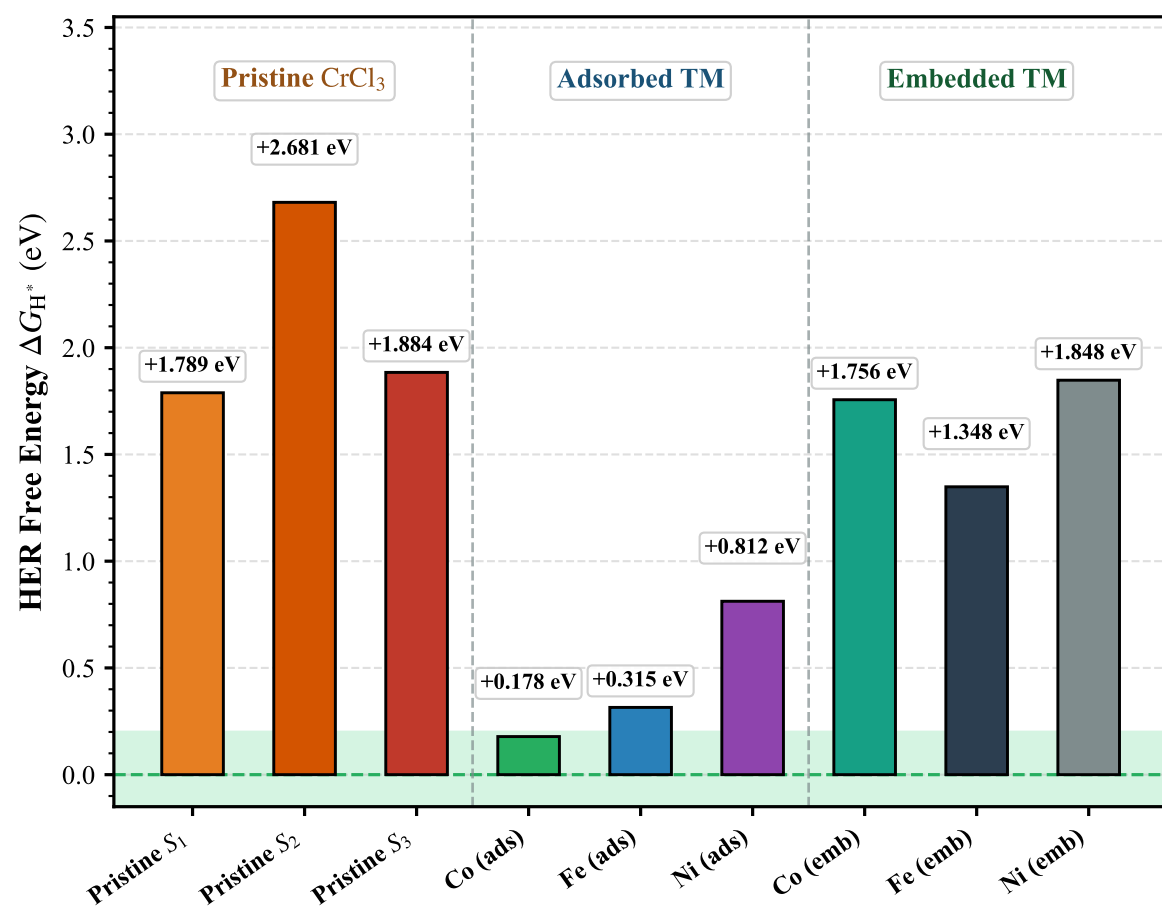


(a) HER Free Energy Diagram with Exact Caique Corrections**(b) Energetic Decomposition: Electronic vs. Vibrational****(c) Calculated HER Free Energy ΔG_{H^*} Across All Systems****(d) Impact of Explicit Vibrational Corrections on ΔG_{H^*}** 